

AIMS Research Internship

Explainable Computer Vision Task

Arjav Jain

25/B01/033

[GitHub Repository](#)

[Drive](#)

Contents

1	Introduction	1
2	Dataset Overview	1
3	Methodology	1
3.1	DenseNet121 Classification Model	2
3.2	Explainability Methods	2
3.2.1	Grad-CAM	2
3.2.2	Integrated Gradients	2
3.3	Qualitative Explainability Analysis	2
3.4	Explainability Evaluation Metrics	2
3.4.1	Faithfulness Metrics	4
3.4.2	Localization Metrics	4
3.4.3	Consistency Metrics	4
3.4.4	Robustness Metrics	5
4	Research Development Process	5
5	Classification Results	6
6	Quantitative Evaluation Results	7
6.1	Training Behaviour	7
6.2	Classification Analysis	8
7	Research Gap Analysis	9
7.1	Faithfulness vs Localization Conflict	9
7.2	Insertion-Deletion Paradox	9
7.3	Low Cross-Method Agreement	9
8	Correlation Analysis	10
8.1	Grad-CAM Correlation Structure	10
9	Proposed Composite Explanation Reliability Score (CERS)	11
10	Discussion	12
11	Conclusion	12
	References	13

1 Introduction

Deep learning models have demonstrated remarkable performance in medical image analysis, particularly for disease detection from chest X-rays. However, despite their high predictive accuracy, these models often operate as “black boxes,” making it difficult to understand the reasoning behind their decisions.

In healthcare applications, model transparency is especially important because clinicians must be able to trust and interpret automated predictions. Explainable Artificial Intelligence (XAI) techniques aim to address this challenge by highlighting image regions that contribute to a model’s decision.

This project investigates the explainability of a DenseNet121 model trained to classify chest X-rays into COVID-19, Normal, and Viral Pneumonia categories. Two widely used XAI methods, Grad-CAM and Integrated Gradients, are evaluated using multiple explainability metrics covering faithfulness, localization, robustness, and consistency.

The study further examines whether commonly used evaluation metrics provide consistent assessments of explanation quality. Based on the observed conflicts between metrics, a Composite Explanation Reliability Score (CERS) framework is proposed to support more balanced and reliable explainability evaluation.

2 Dataset Overview

The study utilizes a chest X-ray dataset consisting of three classes: COVID-19, Normal, and Viral Pneumonia.

A balanced distribution was maintained across classes to minimize bias during training and ensure fair performance assessment.

The dataset was divided into training, validation, and testing subsets. A balanced distribution was maintained across all classes to ensure unbiased model learning and evaluation.

Table 1: Dataset Distribution

Class	Training	Validation	Test
COVID-19	1000	200	145
Normal	1000	200	145
Viral Pneumonia	1000	200	145
Total	3000	600	435

3 Methodology

The overall methodology consists of four major stages:

1. Training a DenseNet121 classification model.
2. Generating explanations using Grad-CAM and Integrated Gradients.
3. Evaluating explanations using multiple explainability metrics.
4. Investigating metric conflicts and proposing a composite evaluation framework.

3.1 DenseNet121 Classification Model

DenseNet121 was selected due to its strong performance in medical imaging applications and its efficient feature reuse through dense connectivity. Transfer learning was employed using ImageNet-pretrained weights.

The final classification layer was modified to perform three-class classification corresponding to COVID-19, Normal, and Viral Pneumonia categories.

Training was performed using categorical cross-entropy loss and the Adam optimizer. Model performance was evaluated using accuracy, precision, recall, and F1-score.

3.2 Explainability Methods

To understand the reasoning behind model predictions, two explainability techniques were implemented.

3.2.1 Grad-CAM

Grad-CAM generates class-specific activation maps using gradients flowing into the final convolutional layer. The resulting heatmaps highlight image regions that contribute most strongly to a given prediction.

3.2.2 Integrated Gradients

Integrated Gradients computes pixel-level attribution scores by integrating gradients from a baseline image to the original input image. Unlike Grad-CAM, Integrated Gradients provides fine-grained pixel-level explanations and satisfies several theoretical attribution properties.

3.3 Qualitative Explainability Analysis

While quantitative metrics provide objective measures of explanation quality, visual inspection remains important for understanding how different explainability methods highlight image regions.

Figure 1 presents representative saliency maps generated using Grad-CAM and Integrated Gradients for correctly classified chest X-ray samples. The visualizations demonstrate noticeable differences in attention patterns between the two methods.

Grad-CAM generally produces smoother and more localized activation regions, whereas Integrated Gradients generates finer pixel-level attribution maps. These qualitative differences motivate the need for multiple evaluation metrics to assess explanation quality from different perspectives.

3.4 Explainability Evaluation Metrics

The quality of an explanation cannot be assessed using a single metric because different metrics evaluate different properties of explanation behaviour. Therefore, explanations generated by Grad-CAM and Integrated Gradients were evaluated across four complementary dimensions: faithfulness, localization, consistency, and robustness.

Saliency Map Comparison: Grad-CAM vs Integrated Gradients (DenseNet121)

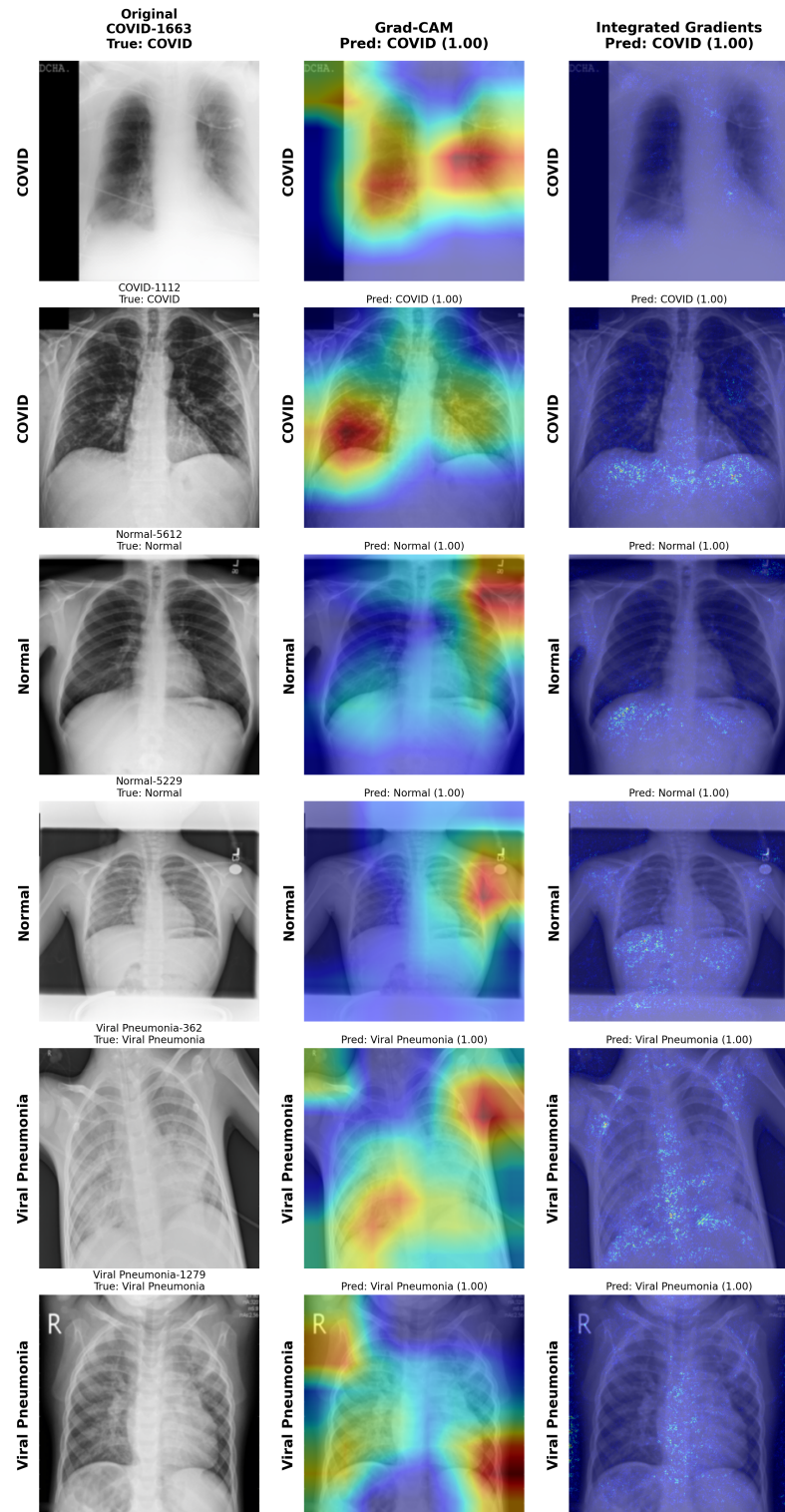


Figure 1: Comparison of saliency visualizations generated by Grad-CAM and Integrated Gradients for representative chest X-ray samples.

3.4.1 Faithfulness Metrics

Faithfulness measures whether the regions highlighted by an explanation genuinely influence the model’s prediction. A faithful explanation should identify image regions that are actually used by the model during decision making.

a) Area Over the Perturbation Curve (AOPC)

AOPC measures the average reduction in model confidence when the most important pixels identified by an explanation are progressively removed. Higher values indicate stronger faithfulness.

In this study, AOPC was used to evaluate whether highlighted chest X-ray regions genuinely influenced the model’s prediction. However, the metric depends on the perturbation strategy and may generate unrealistic modified images.

b) Insertion AUC

Insertion AUC measures how quickly model confidence increases as important pixels are gradually revealed. Higher values indicate that the explanation correctly identifies evidence supporting the prediction.

For this project, the metric assessed how effectively each explanation method captured discriminative image features. Results can vary depending on the chosen baseline image.

c) Deletion AUC

Deletion AUC measures the reduction in prediction confidence when important pixels are removed. Lower scores indicate stronger explanations.

In this work, Deletion AUC was used to determine whether highlighted regions were critical for classification. Similar to AOPC, the metric may be affected by unrealistic perturbations.

3.4.2 Localization Metrics

Faithfulness alone does not guarantee that explanations correspond to clinically meaningful regions. Localization metrics evaluate whether highlighted regions align with relevant anatomical structures.

a) Entropy

Entropy measures the spatial spread of saliency values. Lower entropy indicates concentrated explanations, whereas higher entropy suggests more distributed attention.

In this study, entropy was used to assess explanation sharpness. However, a highly focused explanation is not necessarily clinically meaningful.

b) Lung Focus Score (LFS)

LFS measures the proportion of explanation intensity that falls within the lung region.

The metric was used to evaluate anatomical plausibility in chest X-rays. However, it depends on the quality of the lung mask and does not directly assess faithfulness.

3.4.3 Consistency Metrics

Consistency evaluates whether different explanation methods provide similar interpretations for the same prediction.

a) Spearman Concordance

Spearman Concordance measures the rank correlation between saliency maps generated by different explainability methods. Higher values indicate stronger agreement in the relative importance assigned to image regions.

In this study, Spearman Concordance was used to quantify the consistency between Grad-CAM and Integrated Gradients. Low values suggest that different methods may provide substantially different explanations for the same prediction.

b) Top-k Overlap

Top-k Overlap measures the proportion of highly important pixels shared between two saliency maps. Higher scores indicate stronger spatial agreement.

For this project, Top-k Overlap was used to compare whether Grad-CAM and Integrated Gradients focused on similar chest X-ray regions. However, the metric only evaluates the most salient regions and ignores differences in the remaining attribution map.

3.4.4 Robustness Metrics

A reliable explanation should remain stable under small perturbations that do not significantly alter the underlying image content.

a) Noise Robustness

Noise robustness measures explanation stability when Gaussian noise is added to the input image.

b) Brightness Robustness

Brightness robustness evaluates explanation consistency under intensity variations that commonly occur during image acquisition.

c) Rotation Robustness

Rotation robustness measures explanation stability under small angular transformations.

These robustness metrics were included to ensure that explanation quality was not solely dependent on a specific image representation. Stable explanations are generally more trustworthy in practical clinical settings.

4 Research Development Process

The project was developed iteratively, with each stage motivated by limitations observed in the previous stage.

Initially, a DenseNet121 classification model was trained to establish strong predictive performance. Explainability techniques were then incorporated to interpret model predictions. Multiple evaluation metrics were subsequently implemented to assess explanation quality across several dimensions including faithfulness, localization, consistency, and robustness.

Experimental analysis revealed substantial disagreement among commonly used explainability metrics. These observations motivated the development of the Composite Explanation Reliability Score (CERS), a unified framework that combines multiple evaluation dimensions into a single reliability measure.

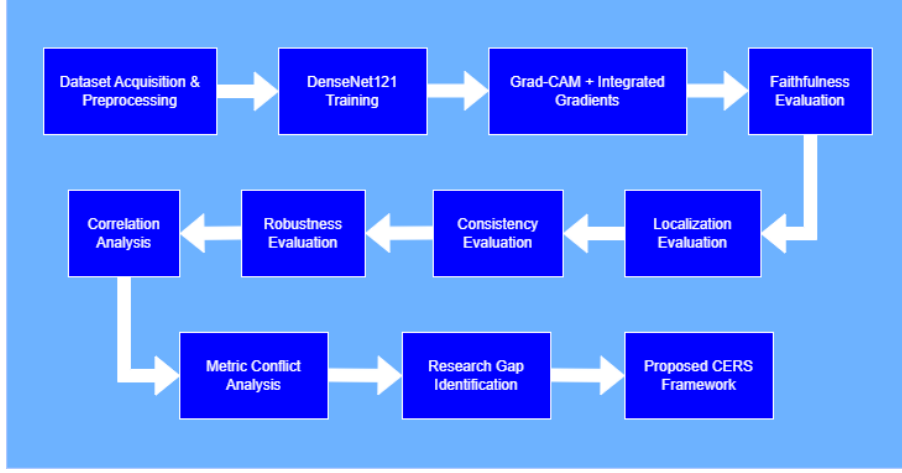


Figure 2: Research workflow illustrating the progressive development of the explainability evaluation framework.

5 Classification Results

The DenseNet121 model achieved excellent classification performance on the test set.

Table 2: Classification Performance

Metric	Score
Accuracy	98.85%
Precision	98.87%
Recall	98.85%
F1 Score	98.85%

These results demonstrate the effectiveness of DenseNet121 for COVID-19 chest X-ray classification and provide a strong foundation for subsequent explainability analysis.

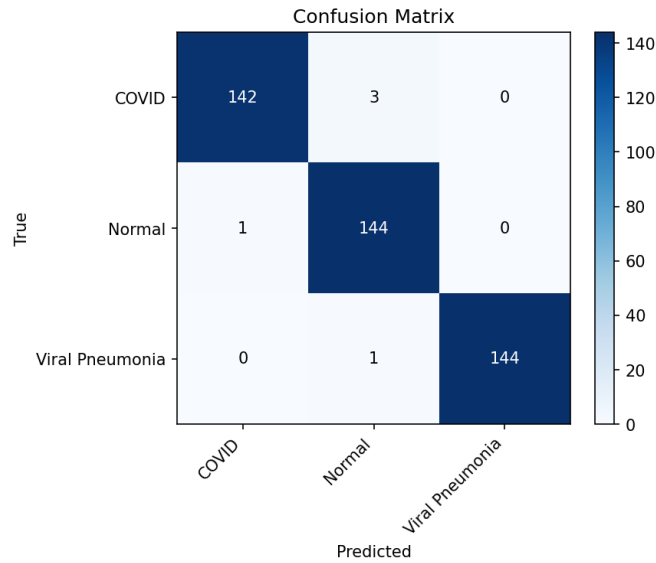


Figure 3: Confusion matrix of DenseNet121 predictions on the test dataset.

6 Quantitative Evaluation Results

Table 3 summarizes the explainability evaluation metrics obtained for Grad-CAM and Integrated Gradients.

Table 3: Explainability Evaluation Results

Metric	Grad-CAM	Integrated Gradients
Entropy	10.7972	10.4957
AOPC	0.4428	0.4904
Insertion AUC	0.7189	0.5124
Deletion AUC	0.5220	0.4719
Lung Focus Score	0.4193	0.2815
Overall Robustness	0.6711	0.5602

The results demonstrate that neither explainability method consistently outperforms the other across all evaluation dimensions.

Integrated Gradients achieved superior AOPC and Deletion AUC scores, suggesting stronger faithfulness under feature removal. In contrast, Grad-CAM obtained substantially higher Insertion AUC, Lung Focus Score, and Robustness scores, indicating better localization and stability.

This disagreement immediately suggests that explanation quality cannot be fully characterized using a single metric.

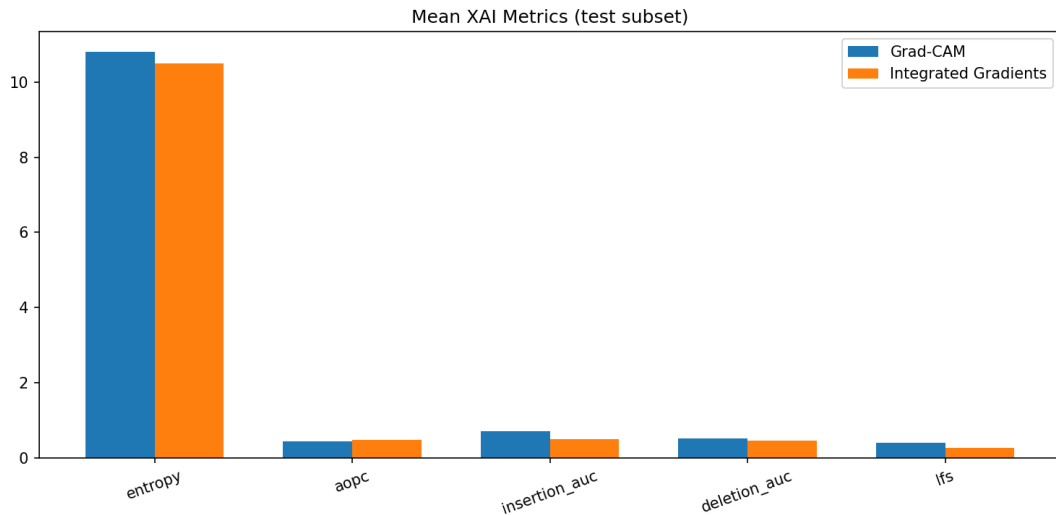


Figure 4: Comparison of explainability metrics for Grad-CAM and Integrated Gradients.

6.1 Training Behaviour

Figure 5 shows the training and validation performance of DenseNet121.

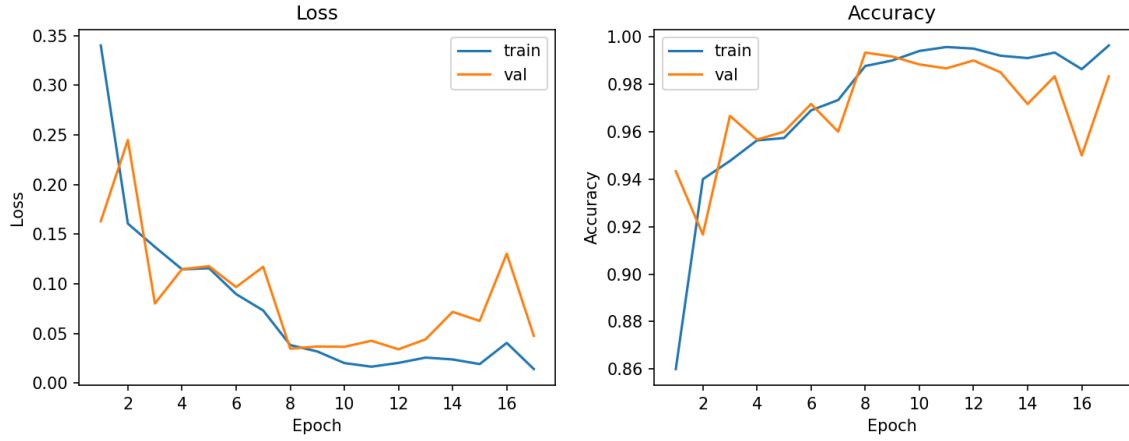


Figure 5: Training and validation loss and accuracy curves.

The model converged rapidly and achieved stable validation performance after approximately twelve epochs. Early stopping prevented overfitting while preserving generalization performance.

6.2 Classification Analysis

The confusion matrix shown in Figure 6 demonstrates strong class-wise performance.

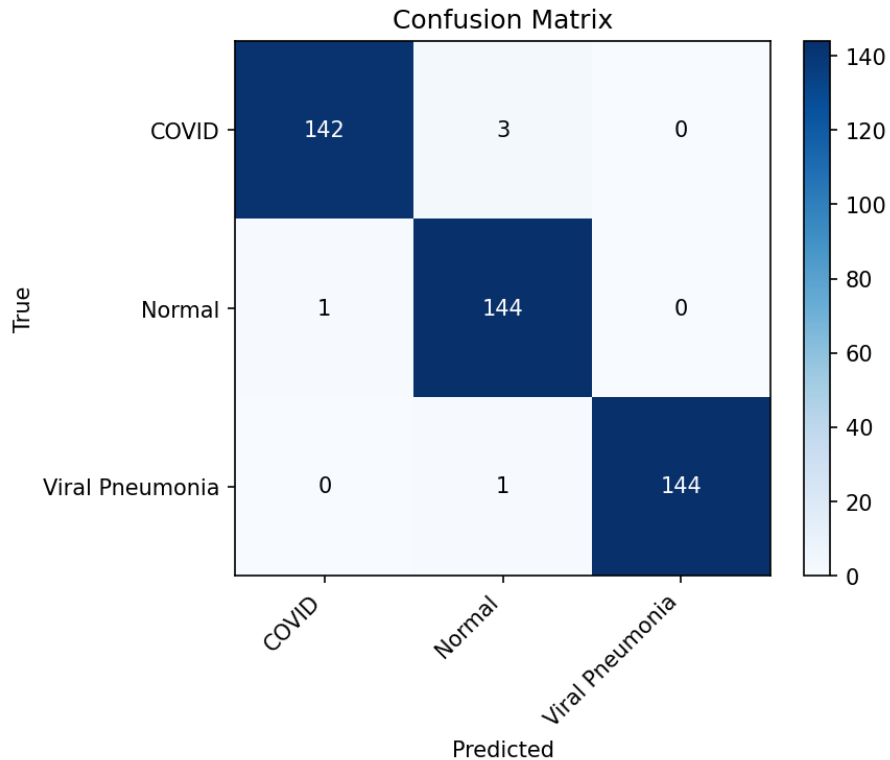


Figure 6: Confusion matrix of DenseNet121 on the test dataset.

Only five misclassifications occurred among 435 test images, yielding an overall accuracy of 98.85%.

7 Research Gap Analysis

One of the primary objectives of this study was to investigate whether commonly used explainability evaluation metrics provide consistent conclusions regarding explanation quality.

The experimental results reveal substantial disagreement among metrics, indicating that current evaluation practices may be insufficient for reliable explanation assessment.

7.1 Faithfulness vs Localization Conflict

Integrated Gradients achieved a higher AOPC score (0.4904) than Grad-CAM (0.4428), suggesting superior faithfulness.

However, Grad-CAM achieved a substantially higher Lung Focus Score (0.4193) compared to Integrated Gradients (0.2815), indicating better anatomical alignment.

This creates a direct conflict:

A method considered more faithful by AOPC is simultaneously considered less clinically plausible by Lung Focus Score.

Therefore, relying on either metric alone would produce contradictory conclusions.

7.2 Insertion-Deletion Paradox

Grad-CAM achieved the highest Insertion AUC:

$$0.7189 > 0.5124$$

while Integrated Gradients achieved the best Deletion AUC:

$$0.4719 < 0.5220$$

Insertion and deletion evaluate opposite perturbation directions.

The fact that each method wins a different metric indicates that explanation rankings depend strongly on the evaluation procedure itself.

This observation highlights a major limitation of current faithfulness metrics.

7.3 Low Cross-Method Agreement

The consistency analysis produced:

- Spearman Concordance = 0.1293
- Top-k Overlap = 0.0736

These values indicate very weak agreement between Grad-CAM and Integrated Gradients.

Despite explaining the same predictions, the two methods frequently highlight different image regions.

Consequently, explanation interpretation may vary significantly depending on the chosen XAI method.

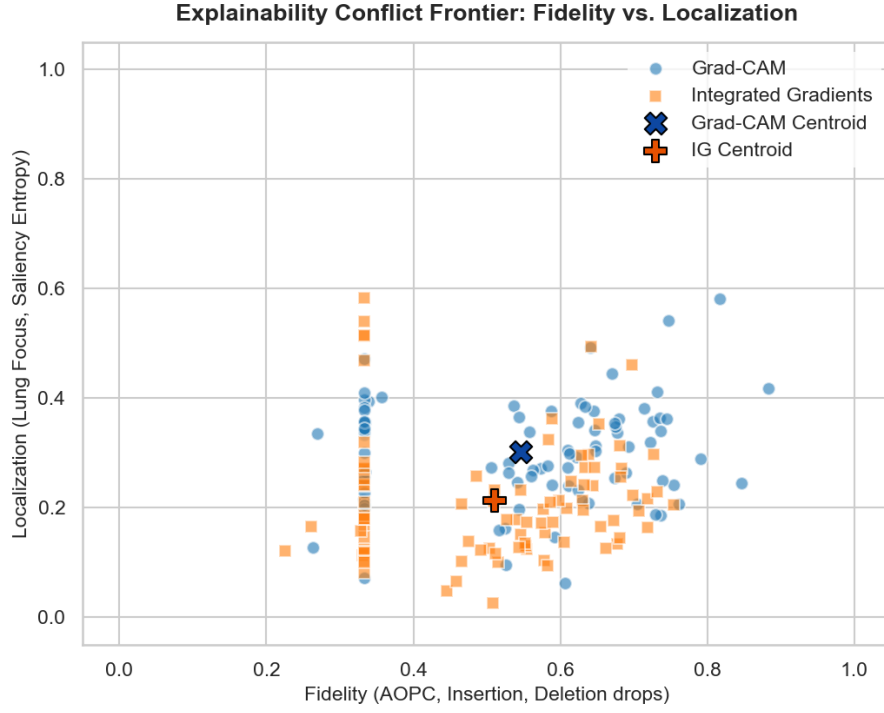
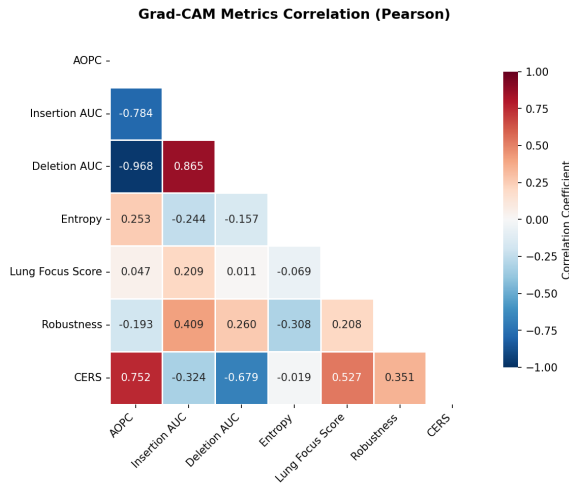


Figure 7: Conflict frontier illustrating the trade-off between fidelity and localization.

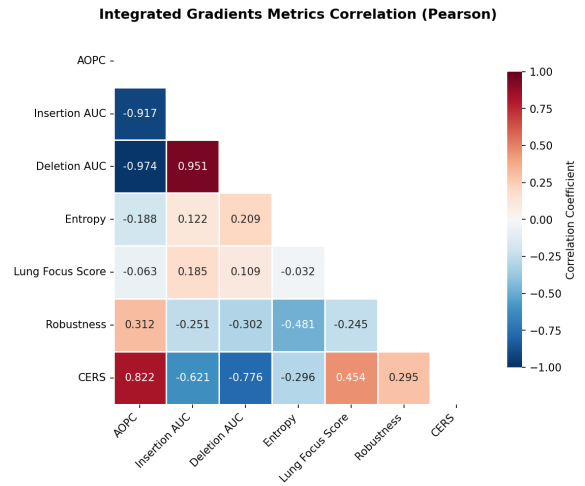
8 Correlation Analysis

To better understand relationships among evaluation metrics, Pearson correlation analysis was performed.

8.1 Grad-CAM Correlation Structure



(a) Grad-CAM



(b) Integrated Gradients

Figure 8: Pearson correlation heatmaps for explainability metrics.

The correlation analysis revealed several important findings.

AOPC and Insertion AUC exhibited strong negative correlations for both methods, confirming that different faithfulness metrics often disagree.

Similarly, Entropy and Lung Focus Score displayed near-zero correlation, indicating that explanation sharpness does not necessarily imply anatomical correctness.

These findings further support the hypothesis that explanation quality is inherently multi-dimensional and cannot be adequately measured using a single metric.

9 Proposed Composite Explanation Reliability Score (CERS)

The experimental results demonstrated that individual metrics frequently produce conflicting rankings.

To address this limitation, a Composite Explanation Reliability Score (CERS) framework is proposed.

The proposed score combines four complementary dimensions:

- Faithfulness
- Localization
- Robustness
- Consistency

The framework can be expressed as

$$CERS = w_f F + w_l L + w_r R + w_c C$$

where

- F = Faithfulness component
- L = Localization component
- R = Robustness component
- C = Consistency component

The proposed framework was motivated by the observation that individual metrics frequently produced conflicting rankings. Instead of relying on a single evaluation criterion, CERS aggregates complementary dimensions of explanation quality into a unified reliability score.

This enables simultaneous consideration of faithfulness, localization accuracy, robustness under perturbations, and cross-method consistency.

Using the proposed framework, the following mean scores were obtained:

Table 4: CERS Results

Method	Mean CERS
Grad-CAM	0.5104
Integrated Gradients	0.4361

The CERS framework favours Grad-CAM because it balances faithfulness, localization, and robustness more effectively than Integrated Gradients.

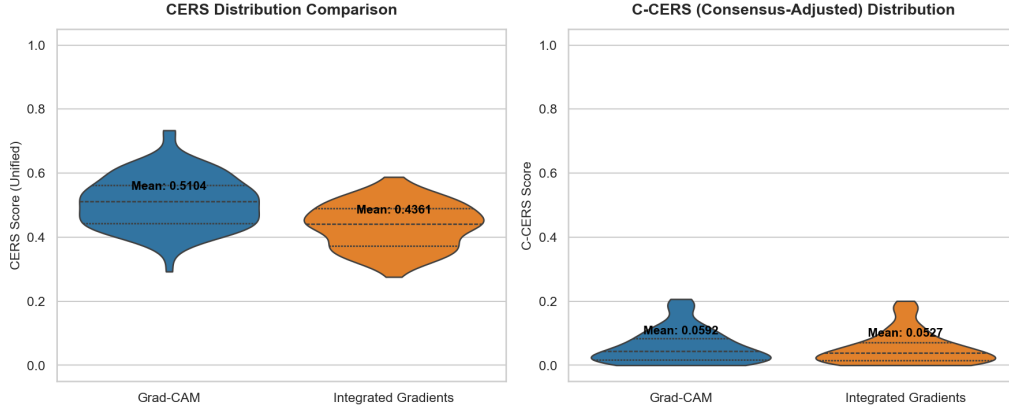


Figure 9: Distribution of CERS and consensus-adjusted CERS scores.

10 Discussion

The results demonstrate that explainability evaluation is fundamentally a multi-objective problem.

Several metrics that are widely used in the literature were found to produce contradictory rankings when applied to the same explanations.

Faithfulness metrics frequently disagreed with localization metrics, while consistency analysis revealed low agreement between explanation methods.

These observations suggest that evaluating explanations using a single metric may produce misleading conclusions.

The proposed CERS framework provides a practical approach for integrating complementary evaluation dimensions into a unified score.

Although the framework does not eliminate all limitations, it offers a more balanced assessment strategy compared to traditional single-metric evaluation approaches.

11 Conclusion

This study developed a DenseNet121-based chest X-ray classification framework for COVID-19, Normal, and Viral Pneumonia detection and achieved a test accuracy of 98.85%.

Grad-CAM and Integrated Gradients were evaluated using faithfulness, localization, robustness, and consistency metrics.

Experimental results revealed substantial disagreement among commonly used explainability evaluation metrics, exposing an important research gap in current XAI assessment methodologies.

To address this limitation, a Composite Explanation Reliability Score (CERS) framework was proposed.

The results demonstrate that explanation quality should be evaluated using multiple complementary perspectives rather than relying on a single metric, particularly in high-stakes medical imaging applications.

References

- [1] Huang, G., Liu, Z., Van Der Maaten, L., and Weinberger, K. Q. *Densely Connected Convolutional Networks*. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2017.
- [2] Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., and Batra, D. *Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization*. Proceedings of the IEEE International Conference on Computer Vision (ICCV), 2017.
- [3] Sundararajan, M., Taly, A., and Yan, Q. *Axiomatic Attribution for Deep Networks*. Proceedings of the International Conference on Machine Learning (ICML), 2017.
- [4] Arrieta, A. B., Díaz-Rodríguez, N., Del Ser, J., Bennetot, A., Tabik, S., Barbado, A., and Herrera, F. *Explainable Artificial Intelligence (XAI): Concepts, Taxonomies, Opportunities and Challenges toward Responsible AI*. Information Fusion, Vol. 58, pp. 82–115, 2020.
- [5] Guidotti, R., Monreale, A., Ruggieri, S., Turini, F., Giannotti, F., and Pedreschi, D. *A Survey of Methods for Explaining Black Box Models*. ACM Computing Surveys, Vol. 51, No. 5, 2019.
- [6] Samek, W., Binder, A., Montavon, G., Lapuschkin, S., and Müller, K. R. *Evaluating the Visualization of What a Deep Neural Network Has Learned*. IEEE Transactions on Neural Networks and Learning Systems, Vol. 28, No. 11, pp. 2660–2673, 2017.
- [7] Petsiuk, V., Das, A., and Saenko, K. *RISE: Randomized Input Sampling for Explanation of Black-box Models*. British Machine Vision Conference (BMVC), 2018.
- [8] Alvarez-Melis, D. and Jaakkola, T. *On the Robustness of Interpretability Methods*. ICML Workshop on Human Interpretability in Machine Learning, 2018.
- [9] Yeh, C. K., Hsieh, C. Y., Suggala, A., Inouye, D., and Ravikumar, P. *On the (In)fidelity and Sensitivity of Explanations*. Advances in Neural Information Processing Systems (NeurIPS), 2019.
- [10] Doshi-Velez, F. and Kim, B. *Towards A Rigorous Science of Interpretable Machine Learning*. arXiv preprint arXiv:1702.08608, 2017.